

# REJIT B ANTONY

## Contact

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## Education

**Vellore Institute of Technology** Chennai, India  
B.Tech – Computer Science Engineering  
July 2023 – Present  
CGPA: **9.12**

**Navdeep Public School** Kollam, Kerala  
High School – CBSE  
May 2021 – May 2023  
Graduated with **94.2%** in Grade 12

## Skills

**Languages**  
Python • Java • JavaScript • LaTeX

**Frameworks & AI**  
Gemini API • PyTorch • TensorFlow •  
React • Node.js • Express.js •  
Scikit-learn • Pandas • NumPy

**Databases**  
MongoDB • SQL • SQLite3

**Tools**  
Gemini CLI / Agents • Git • VS Code •  
JupyterLab • Adobe Photoshop •  
Unity • Godot

**Operating Systems**  
Linux (Arch, WSL, CLI) • Windows

## Certifications

- 1st Place – FOSSIT**  
(LUGVITC, April 2024)
- Volunteer Recognition Award**  
– Raptee.HV, CodeVolt (Mar 2025)
- Full Stack MERN** Certification  
(Delta Course) – ApnaCollege
- Agile & Design Thinking**  
– IBM
- DevOps Fundamentals**  
– IBM

## Projects

**Reinforcement Learning – BipedalWalker (TD3 & TD3-FORK)**  GitHub  
*PyTorch • Gymnasium • CUDA*

- Implemented TD3 from scratch in PyTorch (replay buffer, twin critics, actor, Polyak target updates) with no RL library dependencies; trained on BipedalWalker-v3 to avg reward **+280/300** in **1M steps** on personal GPU.
- Built a live **6-panel** Matplotlib training dashboard (reward curves, Q-value drift, action distributions, TD error) and a threaded pygame render window with HUD overlay displaying training in real time without blocking the training loop.
- Extended the agent with TD3-FORK (forward dynamics model predicting next state from current state-action) to address long-horizon credit assignment failure on BipedalWalkerHardcore-v3; reward improved from **-40 to +100+** within **900k steps**.

**Placement Tracker Platform (Full-Stack Features & Data Pipelines)**  GitHub  
*TypeScript • Svelte • SQLite3*

- Engineered granular placement tracking and shortlisting pipelines using **Svelte** and **SQLite**.
- Architected data import pipelines processing **15,000+** student and company records via **TypeScript** scripts, ensuring robust **data integrity and mapping accuracy**.
- Executed a structural database migration from legacy student IDs to a unified **NeoID** architecture, implementing efficient SQL joins and pagination to slash query latency from **2000ms to 50ms**.

**Image Denoising Autoencoder**  GitHub  
*TensorFlow • Keras • Python*

- Built a hybrid Inception-ResNet denoising autoencoder in TensorFlow/Keras trained on DIV2K, using Inception blocks (**1×1, 3×3, 5×5**) and stacked ResNet residual blocks across a **3-stage** encoder.
- Implemented a U-Net style decoder with skip connections and mixed-precision training (float16) to preserve spatial detail while halving activation VRAM consumption.
- Engineered a memory-efficient pipeline using a custom Keras Sequence generator for on-the-fly noise augmentation across **5000 patches** from **800 source images**.

## Extracurricular Activities

**The Linux Club** VIT Chennai  
*Media Lead* June 2024 – April 2026

- Directed content photography and videography, curating and managing a 100GB+ media repository to drive social media campaigns and boost community engagement at VIT Chennai.
- Designed event websites and promotional posters for club events and open-source initiatives.

**The Robotics Club** VIT Chennai  
*Media & Design Lead* April 2025 – April 2026

- Designed promotional posters, banners, and visual assets for robotics events and competitions.
- Managed event photography and media content, streamlining cloud storage uploads and social media distribution to maximize club visibility and outreach.

**CodeVolt '25 Hackathon** VIT Chennai  
*Volunteer* March 2025

- Supported event coordination, ensuring a seamless experience for participants and judges.
- Assisted teams with technical queries and troubleshooting during the hackathon.
- Collaborated with organizers and volunteers for smooth event execution.